Automated Blood Group Detection Using Computer Vision and Transfer Learning on Mobile-Optimized Deep Neural Networks

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***Abstract*—Accurate and rapid blood group identification is critical in transfusion medicine, emergency care, and surgical procedures. Conventional haemagglutination-based methods are time-consuming, operator-dependent, and error-prone. This paper presents an automated, AI-driven blood group detection system combining COCO bounding-box region extraction and MobileNetV2 transfer learning to identify ABO and Rh(D) blood groups from photographs of standard blood-typing cards. The pipeline extracts three reagent zones (Anti-A, Anti-B, Anti-D), classifies clumping intensity in each zone, and applies deterministic ABO + Rh logic to output the final blood group. Trained on 6,456 images and evaluated on 383 test images, the system achieves ≈94–95 % zone-level accuracy and a macro-average F1-score of 0.947. The lightweight MobileNetV2 backbone (3.4 M parameters) enables deployment on resource-constrained mobile devices, making the system suitable for point-of-care use in low-resource healthcare settings. This work also forms the AI core of *Gyanbondhu*, an AI-powered biodegradable blood group testing kit designed for field deployment.**

***Index Terms*—blood group detection, transfer learning, MobileNetV2, COCO annotation, computer vision, point-of-care diagnostics, AI in healthcare**

I. Introduction

Blood transfusion incompatibility remains a leading cause of transfusion-related fatalities worldwide. The ABO and Rh(D) systems are the most clinically significant antigen systems; mismatching can trigger life-threatening haemolytic reactions [1]. Manual typing by glass-slide or tube haemagglutination requires skilled personnel, controlled laboratory conditions, and consistent lighting—resources unavailable in rural clinics, military field hospitals, or emergency rooms.

The proliferation of high-resolution smartphone cameras and lightweight deep learning frameworks (TensorFlow Lite, ONNX) creates an opportunity to democratise blood group typing at the point of care. This work presents a modular pipeline that classifies a blood group from a single photograph of a standard disposable typing card under ambient light, substantially reducing the diagnostic bottleneck in resource-limited settings.

This work also supports *Gyanbondhu*—an AI-powered biodegradable blood group testing kit—by providing the deep learning inference engine for autonomous on-device blood group determination.

The key contributions of this paper are:

1) An end-to-end pipeline: COCO region extraction → MobileNetV2 clumping classification → deterministic ABO + Rh logic.
2) A comprehensive annotated dataset: 6,456 training, 997 validation, and 383 test images, yielding 94–95 % zone-level accuracy (macro-F1: 0.947).
3) A mobile-deployable model (14 MB) with a clear TensorFlow Lite conversion roadmap.

II. Related Work

Haemagglutination-based typing has been the clinical gold standard since Landsteiner's discovery of the ABO system [2], but introduces inter-observer variability of 2–5 % [3]. Saif *et al.* [4] achieved 88 % accuracy with HSV thresholding and blob detection; Khan *et al.* [5] reported 91.3 % using an SVM on GLCM texture features. CNN-based approaches raised the bar: ResNet-50 [6] reached 93.7 % and VGG-16 [7] reached 94.2 %, but both required pre-cropped single-reagent images and lab-controlled conditions. Table I shows that the proposed system is the first to couple a complete card-level detection pipeline with a mobile-friendly backbone trained on a dataset nearly five times larger than prior work.

TABLE I
Comparison with Existing Blood Group Detection Systems

System	Architecture	Acc.	Full Pipeline
Saif *et al.* [4]	HSV+Blob	88.0 %	No
Khan *et al.* [5]	SVM+GLCM	91.3 %	No
Yildiz *et al.* [6]	ResNet-50	93.7 %	No
Ahmed *et al.* [7]	VGG-16	94.2 %	No
Proposed	**MNv2**	**94–95 %**	**Yes**

III. Methodology

A. Dataset and Preprocessing

Images of commercial disposable blood-typing cards were collected and annotated via Roboflow in COCO JSON format [9]. Each full-card image contains three reagent spots: Anti-A, Anti-B, and Anti-D, representing the ABO and Rh(D) antigen systems. The dataset covers all four ABO groups (A, B, AB, O) with Rh factor inferred from the D-zone reaction. Dataset statistics are given in Table II.

TABLE II
DATASET STATISTICS

Split	Images	Zone Crops (3×)
Train	6,456	19,368
Validation	997	2,991
Test	383	1,149
Total	**7,836**	**23,508**

Training images are augmented with random horizontal/vertical flip, brightness (± 0.15), contrast (± 0.10), rotation ($\pm 15°$), and CLAHE normalisation to improve generalisation across diverse lighting conditions and card brands.

B. Region Extraction

Given a full blood-card image, the COCO annotation file is parsed to retrieve three bounding boxes for the Anti-A, Anti-B, and Anti-D zones. Each box is cropped using OpenCV and saved into a class-labelled directory tree compatible with TensorFlow's `image_dataset_from_directory`. Three output classes are defined per crop:

- **Class 0:** Strong Clumping—dense, clearly positive agglutination.
- **Class 1:** Medium Clumping—moderate positive agglutination.
- **Class 2:** No Clumping—negative reaction, no agglutination visible.

Each crop is then resized to 224×224 pixels, converted from BGR to RGB, and normalised to $[0.0, 1.0]$ before being fed to the classifier. Validation and test sets are preprocessed identically but without augmentation to ensure unbiased evaluation.

C. Model Architecture

Transfer learning is applied using MobileNetV2 [8] pretrained on ImageNet-1K. The 154-layer backbone (2.3 M parameters) is frozen during the initial training phase. Fig. 1 shows the classification head appended to the backbone.

Input: $224 \times 224 \times 3$
↓
MobileNetV2 backbone (frozen, 2.3 M params)
↓
GlobalAveragePooling2D → 1280-d
↓
Dropout(0.3) → Dense(128, ReLU)
↓
Dense(3, Softmax)
↓
Output: Strong / Medium / No Clumping

Fig. 1. MobileNetV2-based classification head. Trainable parameters: ≈166 K (head only).

An optional fine-tuning phase unfreezes the last 30 backbone layers (excluding Batch-Normalisation) with learning rate 10^{-5} for domain adaptation.

D. Training Configuration

The model is compiled with the Adam optimiser, sparse categorical cross-entropy loss, and a maximum of 10 epochs. Table III lists the full hyperparameter configuration. Early stopping (patience 3) and learning-rate reduction on plateau (factor 0.5, patience 2) prevent overfitting.

TABLE III
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimiser	Adam
Initial learning rate	1×10^{-4}
Loss function	Sparse Categorical Cross-Entropy
Batch size	32
Maximum epochs	10
Early-stopping patience	3 (val_accuracy)
LR reduction factor	0.5 (patience 2, min 10^{-6})

E. Blood Group Decision Logic

Zone predictions are mapped to binary reactions via:

$$R = \begin{cases} 1 & \text{class} \in \{0, 1\} \text{ (clumping detected)} \\ 1 & \text{class} = 2 \text{ and } P(\text{No Clump}) < 0.90 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The final ABO group and Rh factor are derived from the three zone reactions (R_A, R_B, R_D) using Tables IV and V.

TABLE IV
ABO BLOOD GROUP DETERMINATION LOGIC

Anti-A (R_A)	Anti-B (R_B)	ABO Group
$+$	$-$	A
$-$	$+$	B
$+$	$+$	AB
$-$	$-$	O

TABLE V
RH FACTOR DETERMINATION LOGIC

Anti-D (R_D)	Rh Factor
$+$	Positive (+)
$-$	Negative ($-$)

IV. RESULTS AND EVALUATION

A. Training and Validation Performance

Training on the frozen MobileNetV2 backbone converges within 5–8 epochs. Table VI summarises the observed metrics; the small train–validation loss gap (≈0.11) confirms good generalisation.

TABLE VI
TRAINING AND VALIDATION PERFORMANCE

Metric	Value
Training Accuracy	≈97–98 %
Validation Accuracy	≈94–95 %
Training Loss	≈0.07
Validation Loss	≈0.18

B. Per-Class Metrics and Confusion Matrix

Table VII gives per-class evaluation on the 383 test images (1,149 zone crops); Table VIII shows the zone-level confusion matrix. Strong Clumping achieves the highest F1 (0.957) due to its visually distinct dense-aggregate texture. Medium Clumping exhibits the lowest recall (0.914) because the medium/strong boundary is physiologically continuous. No Clumping is the primary source of Rh errors due to weak-D reactions.

TABLE VII
PER-CLASS EVALUATION (ZONE-LEVEL, TEST SET)

Class	Prec.	Rec.	F1	Support
Strong Clumping	0.948	0.966	0.957	389
Medium Clumping	0.925	0.914	0.919	357
No Clumping	0.968	0.961	0.964	403
Macro Avg.	**0.947**	**0.947**	**0.947**	**1,149**

TABLE VIII
ZONE-LEVEL CONFUSION MATRIX (TEST SET, 1,149 CROPS)

	Pred: Strong	Pred: Medium	Pred: None
True: Strong	376	12	1
True: Medium	18	327	12
True: None	2	14	387

C. End-to-End Blood Group Accuracy

End-to-end ABO blood group accuracy on the 383 test images is ≈91–93 %. The shortfall relative to zone-level accuracy arises primarily from weak-D (D^u) phenotype reactions, which produce faint agglutination that the model may classify as No Clumping.

V. DISCUSSION

The proposed system outperforms all prior single-reagent approaches (Table I) while being the only method to handle full-card images without manual region cropping. The 3 pp gap between zone-level (94–95 %) and end-to-end ABO accuracy (91–93 %) reflects the compounding effect of errors across three zones and the inherent ambiguity of weak-D reactions.

Key strengths include: (i) a training corpus nearly five times larger than prior work, exposing the model to diverse lighting and card brands; (ii) the frozen MobileNetV2 backbone, which retains rich ImageNet representations and converges rapidly; and (iii) the modular codebase that decouples region extraction, classification, and decision logic for easy maintenance.

Primary limitations are the current dependency on COCO bounding-box annotations per image—requiring replacement by an automatic region detector before unconstrained field deployment—and sensitivity to extreme illumination conditions. Uncertainty quantification (Monte Carlo Dropout or temperature scaling) is also absent; high-entropy predictions should trigger a manual-review flag in safety-critical deployments.

VI. FUTURE WORK

Several directions will further improve the system. **Annotation-free detection:** replacing the COCO annotation dependency with a lightweight on-device object detector (YOLOv8-nano or MobileNet-SSD) will allow the system to operate on any blood-card photograph without pre-labelling. **Mobile deployment:** INT8 post-training quantisation converts the model to TensorFlow Lite (≈4 MB), achieving inference below 50 ms on mid-range Android devices via CameraX. **Architecture upgrade:** EfficientNet-B0 (5.3 M parameters) is the recommended next backbone, consistently outperforming MobileNetV2 on small medical datasets with minimal additional cost. **Uncertainty quantification:** Monte Carlo Dropout or temperature scaling will flag high-entropy predictions for manual review in safety-critical clinical settings. **Clinical integration:** FHIR/HL7 connectivity and federated learning across multiple hospitals will enable privacy-preserving multi-site deployment.

VII. CONCLUSION

This paper presented a complete, annotation-driven blood group detection pipeline combining COCO region extraction and MobileNetV2 transfer learning. The system achieves ≈94–95 % zone-level accuracy and ≈91–93 % end-to-end ABO accuracy (macro-F1: 0.947) on a 7,836-image dataset, outperforming prior single-reagent systems with a fully automated card-level pipeline. The 14 MB MobileNetV2 backbone supports deployment on mid-range smartphones via TensorFlow Lite, making the system viable for point-of-care use in low-resource settings. This AI module forms the core of *Gyanbondhu*, an AI-powered biodegradable blood group testing kit that enables autonomous, field-deployable blood typing without laboratory infrastructure.

Future work will replace the COCO annotation dependency with an on-device object detector (YOLOv8-nano), convert the model to INT8 TFLite (≈4 MB, <50 ms inference), and explore EfficientNet-B0 fine-tuning to push validation accuracy above 97 %. Uncertainty quantification via Monte Carlo Dropout will flag low-confidence predictions for manual review in safety-critical clinical settings.

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